Critical Infrastructure Security

IY5612 (IY3612/IY4612) — Academic Year 2025/2026 (Unit 4)

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Unit 4— Economic Interdependency Models

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Part III

Economic Models of Interdependency

Critical Infrastructure Interdependency

Economic Models

This block describes the concepts of interdependency and high-level economic models of interdependencies among critical infrastructures, emphasising the role of information and communication technology sectors that are the main cyber security drivers:

- Introduction to modelling approaches for dependencies and interdependencies
- Input-Output economic models
- Case Study: Centrality measures for the analysis of financial networks

Recommended Reading

- N. K. Svendsen, S. Wolthusen:
  Modelling Approaches
  In: J. Lopez, R. Setola, S. Wolthusen (eds.):
  Critical Infrastructure Protection: Information Infrastructure Models, Analysis, and Defense
  Lecture Notes in Computer Science Vol. 7130, Springer-Verlag, 2012
- Y.Y. Haimes: Risk Modelling, Assessment, and Management, 3rd ed., Wiley, 2009
- S. Battiston, M. Puglia, R. Kaushik, P. Tasca, G. Caldarelli:
  DebtRank: Too Central to Fail? Financial Networks, the FED and Systemic Risk
  Nature Scientific Reports **2**(541):1-6 (Aug. 2012)

Dependency and Interdependency

Understanding Dependency Models

Thus far we have concentrated mainly on understanding the **structure** of networks and the development of models which allow us to characterise this structure

- Such models must necessarily simplify, permitting the study of structures that would otherwise not be visible or accessible either mathematically or in some form of algorithmic or simulative approach

> The best material model of a cat is another, or preferably the same, cat
> N. Wiener

One important question arising naturally is the **dependency** of nodes on one another in such networks:

- A successful attack would intuitively affect dependent nodes, determining relative need for protection

Dependency and Interdependency
Model Properties

In addition to the global properties studied so far, it is often of interest to identify **dependencies**:

- Where relations are captured e.g. by a (directed) graph, determining dependent nodes is trivial provided only simple transitive properties are assumed
- Determining the set of nodes on which a node depends is also straightforward in a directed graph
- Where the dependency graph has cycles, we obtain **interdependencies**

The above holds for simple graphs, but it may often be necessary to include loops or to introduce weights to represent graph properties

Modelling Infrastructures
Model Hierarchies (1/3)

In models described previously, considerations so far have been restricted to the same abstraction level

- Not always possible, and different abstraction levels will need to be considered depending on information available and results sought:

Abstraction Level	**Modelled Entities** **Model Actors** **Modelling Method Scope**
National Security	Infrastructure sectors, state or supranational entities, general public Policy-makers, defence, and military organisations including national and international entities Qualitative analyses and macro-economic models

Modelling Infrastructures
Model Hierarchies (2/3)

Abstraction Level	Modelled Entities Model Actors Modelling Method Scope
Legal	Infrastructure sectors and intra-sector actors, external entities
	Law enforcement, intelligence, and regulatory agencies
	Qualitative analyses and (qualitative) economic models
Technical	Trans-sector and sector-specific technical systems
	Technical systems, codified policies and mechanisms, management
	Qualitative and high-level quantitative models

Modelling Infrastructures
Model Hierarchies (3/3)

Abstraction Level	**Modelled Entities** **Model Actors** **Modelling Method Scope**
System	Smaller-scale technical and organisational systems and their interactions, also across infrastructure sectors Management, engineers, and technicians Qualitative, but also quantitative models with substantial technical and organisational fidelity

Modelling Scales for CI Networks (1/2)

The national security and legal levels in the hierarchy outlined are largely beyond the scope of what can be modelled with the methods studied in this module

A number of parameters influence what can be modelled:

- Understanding of modelling object (e.g. suitable mathematical models)
- Availability of quantitative data: Models may be highly sensitive to choice of parameters
- Where such models are used for prediction, simulations are often required — this limits the granularity and complexity of models
  - Granularity can be the *level of detail* or *scale* of a model
  - Orthogonally, the *time-scale* must often be considered

Modelling Scales for CI Networks (2)

Models will be designed to capture different scales, rarely encompassing more than one or two levels:

Scale	Definition
Part	Smallest component of a system identifiable in an analysis
Unit	Functionally related set of parts
Subsystem	Functionally related set of units
System	Grouping of subsystems
Infrastructure	Self-contained set of similar systems
Interdependent Infrastructures	National and supranational interconnected infrastructures

(Taxonomy of modelling scales according to Rinaldi *et al.* IEEE Control Systems, Vol. 21, No. 6, pp. 11-25)

Input-Output Models

At a relatively high level, **economic dependencies** can be of interest, ranging from CI sectors in a region or entire country or within a single organisation

This is one of the earliest models for studying interdependencies, originally developed as Input-Output Models by Leontief in the 1930s (Nobel prize in Economics 1973)

- Relations among industry sectors were captured as inputs and outputs in a matrix, yielding a set of linear equations
- In the simplest case each sector i produces one output x_i, each unit requires a_{ij} units from sector j
- Sectors are interdependent, producing and consuming outputs mutually
  - Note that these models do not normally capture nested I/O as may be relevant for supply chain analyses

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Input-Output models were used widely in the Soviet Union for economic planning . . .

Leontief Input-Output Models

In this formulation, the output of a sector (or other aggregate) is represented as

$$x_i = a_{i1}x_1 + a_{i2}x_2 + \ldots + a_{in}x_n + d_i$$

which also makes it easy to to determine the entire system's state in the matrix as well as the reqired output for a desired demand vector

This model has been extended by Haimes and Jiang in the form of an **Inoperability Input-Output** model (IIM), capturing the level of dysfunction relative to its nominal state

- This basic model has been extended multiple times, e.g. to also capture dynamic effects such as recovery from faults
- IIM/DIIM are not limited to cross-sectoral models and have been used in e.g. information systems

Input-Output Models
Limitations of Input-Output Models

IIM/DIIM are mainly useful to understand the impact of failures, and are not specific to resources (physical, information, etc.)

- Generally only a single, homogeneous output per sector is considered, implying a minimum level of abstraction
- These models are also of limited use in capturing malicious effects as even recovery is not captured as an objective-driven mechanism
- For capturing deliberate attacks such as disruptions caused by cyber attacks, both steady-state and cascading effects are of interest

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Several further limitations stem from the need to know the exact relationships between inputs and outputs produced, having to rely on a linear model, and only a single metric to optimise for — multiple contradicting objectives may exist but cannot be captured directly

Input-Output Models

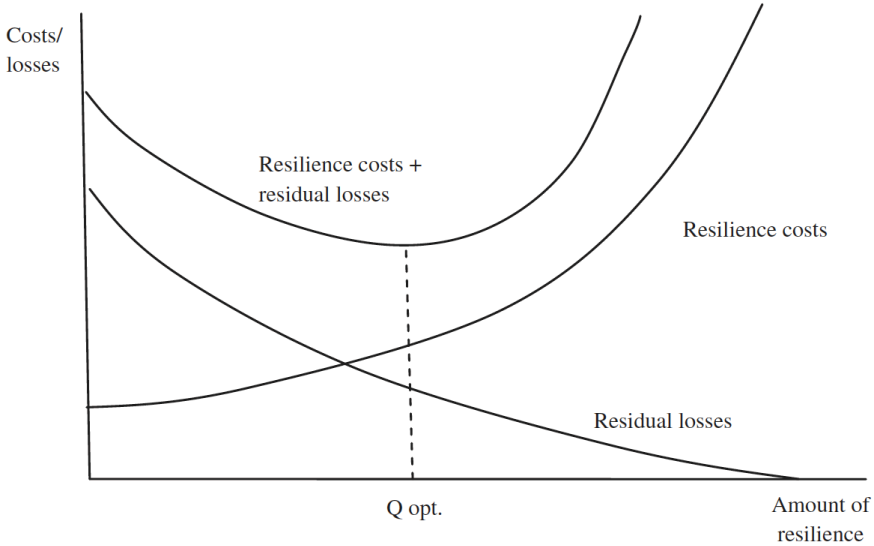
Extending and Applying Input-Output Models

Dynamic extensions to Input-Output models have been developed since the 1960s, more commonly in discrete but also time-varying variants

Although mostly qualitative, such models are useful for understanding risk and impact of interdependencies

- This allows characterisation of resilience and constraints on recovery from incidents
- Other extensions include proposals by Barker and Santos include *inventories* in the model to capture the ability to buffer temporary inoperability

Cost of Resilience



Steady-State Graph Models

As in the case of the basic IIM, the graph models considered thus far are **static** in nature, i.e. they do not consider any perturbation such as will arise from a natural disaster or particularly a deliberate attack:

- Despite this important limitation, knowledge of the network structure itself can give important insights into potential vulnerabilities

The networks studied are not limited to actual networks, but can e.g. be arbitrary dependencies

- The precise mechanism may not be necessary to capture the dependency or interdependency one is interested in
- An example of a metric identifying the criticality of a resource is the **centrality** of a node

Dependencies and interdependencies are difficult to spot and often non-obvious in cyber-(physical) systems, but not just there. . .

Dependency Graph Models

Example: Financial Networks and Centrality

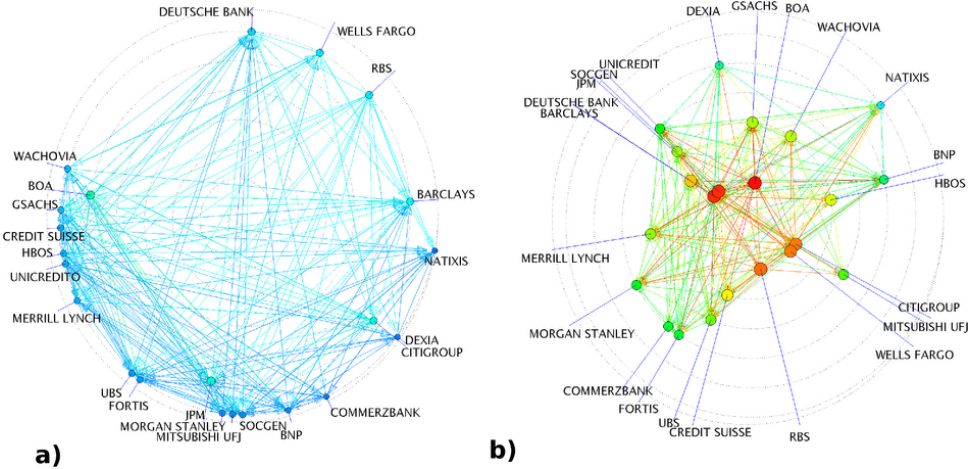
Several variants of the centrality measure for graphs introduced previously exist, including the basic page rank metric and algorithm used by the Google search engine

Battiston *et al.* introduced an analogous centrality metric (DebtRank) to understand the relative importance of financial institutions

- This identifies institutions that are the largest borrowers — also relative to asset size — but more importantly also their interconnection with other institutions
- This interconnection is critical as institutions are cross-linked by transactions, holding equity, and debt of one another
- Analyses such as these can help to understand how events propagate across networks

Dependency Graph Models

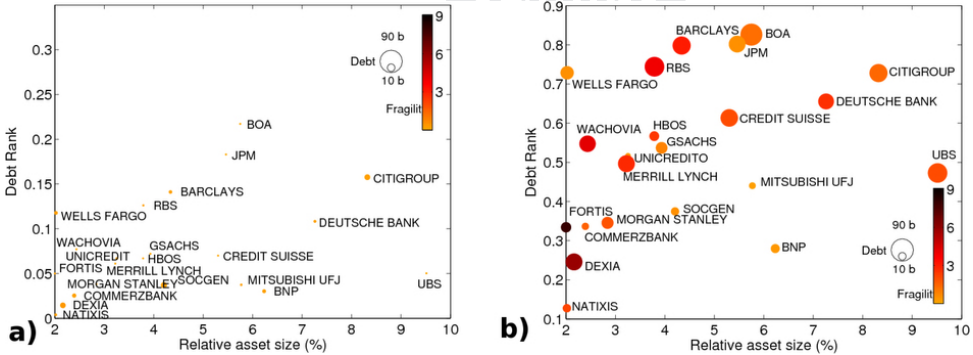
Fragility in Financial Networks (1)



Network of top borrowers (Battiston *et al.*, 2012)
(a) Beginning of financial crisis (Aug. 2007)
(b) Height of crisis (Mar. 2009)

Dependency Graph Models
Fragility in Financial Networks (2)

Relative asset size (core tier 1 ratio) is an insufficient metric for fragility — the fragility of both Deutsche Bank and UBS was higher than one would have expected based on capital ratios alone:



Asset size and fragility (Battiston *et al.*, 2012)
(a) Beginning of financial crisis (Aug. 2007)
(b) Height of crisis (Mar. 2009)

Dependency Models
Purpose and Limitations

The previous example considered a graph model of a single infrastructure and very abstract relations among entities

- Particularly in cyber-physical systems a more in-depth view of connections is required
- Typically multiple components are linked by computer and communication networks, hence requiring an understanding of network topology and both static and dynamic effects
- Most systems of interest for cyber security may not only consist of a substantial number of interacting components, but also will be dynamic, with the topology and/or types of interactions (dependencies) changing over time

Dependency models can help answer the key question of asset criticality and which assets need to be protected — but **not** what attacker objectives might be

Summary

This section provided material on

- Dependencies and Interdependencies
- Modelling Abstractions
- Critical Infrastructure Modelling Approaches
- Economic Models of Interdependency